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Semiconductor device and semiconductor device testing method

Abstract

A semiconductor device includes: a first semiconductor chip including a first internal circuit, first flip-flop circuits connected to the first internal circuit, first selectors, and first electrodes connected to an output of the first selector; first connection conductors; and a second semiconductor chip including second electrodes connected to the first electrode via the first connection conductor and a second internal circuit connected to at least one second electrode. At least one of the first and second semiconductor chips includes a test circuit. The test circuit includes a first detection circuit receiving a signal from each second electrode, a first selector control circuit controlling the selection of the first selector, and an expected value generation circuit. Each first selector includes a signal input receiving a signal from at least one first flip-flop circuit and an expected value input receiving an expected value signal from the expected value generation circuit.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2016/0131704	12/2015	Kawoosa	714/729	G11C 29/32
2021/0231732	12/2020	Park	N/A	G01R 31/3177
2022/0341991	12/2021	Lai	N/A	G11C 29/56

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2007303897	12/2006	JP	N/A

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION
(1) This application claims priority under 35 USC 119 from Japanese Patent application No. 2022-143300 filed on Sep. 8, 2022, the disclosure of which is incorporated by reference herein.
BACKGROUND
Technical Field
(2) The disclosure relates to a semiconductor device and a semiconductor device testing method.
Description of Related Art
(3) Japanese Patent Application Laid-Open No. 2007-303897 discloses a scan test method for multi-chip packages. In this method, data sent out from a first scan chain circuit is passed to a second scan chain circuit. The data received by the second scan chain circuit is compared with expected values. Based on the comparison of expected values, the success or failure of data transmission/reception is determined.
(4) When mounting a plurality of semiconductor chips (Large Scale Integrated Circuit: LSI) in three dimensions (3D) or a multi-chip package (MCP), it is also necessary to test whether or not connections are proper between mounted semiconductor chips in addition to the test of the

individual semiconductor chip.
(5) The 3D mounting is performed in various ways by connections via wires or bumps and the use of through wires. Any form requires the test of the interconnection between the semiconductor chips.
(6) In testability (DFT), the interconnections between two semiconductor chips are placed in a series of shift registers to form a scan chain. A pattern is inputted from the input to the scan chain formed in this way and an output pattern is obtained from the output.
(7) However, since the path of the test pattern includes scan chains, information on a failure interconnection at a location between semiconductor chips reaches the output after passing through elements or circuits on the downstream side of that interconnection. The test time increases when passing through the elements or circuits on the downstream side.
(8) The disclosure provides a semiconductor device and a semiconductor device testing method capable of shortening the time for testing whether the connection between semiconductor chips is good or bad.
SUMMARY
(9) An aspect of the disclosure provides a semiconductor device. The semiconductor device includes: a first semiconductor chip which includes a first internal circuit, multiple first flip-flop circuits connected to the first internal circuit, multiple first selectors, and multiple first electrodes connected to respective outputs of the first selectors; multiple first connection conductors; and a second semiconductor chip which includes multiple second electrodes respectively connected to the first electrodes via the first connection conductors and a second internal circuit connected to at least one of the second electrodes. At least one of the first semiconductor chip and the second semiconductor chip includes a part of a test circuit unit. The test circuit unit includes a first detection circuit which receives a signal from each of the second electrodes, a first selector control circuit which controls the first selectors, a first expected value generation circuit which generates a first expected value signal including a first expected value, and a test circuit which receives an output of the first detection circuit. Each of the first selectors includes a first signal input which receives a signal from any one of the first flip-flop circuits and a first expected value input which receives the first expected value signal from the first expected value generation circuit.
(10) Another aspect of the disclosure provides a semiconductor device. The semiconductor device includes: a first semiconductor chip which includes a first internal circuit, multiple first flip-flop circuits connected to the first internal circuit, multiple first selectors, and multiple first electrodes connected to respective outputs of the first selectors; multiple first connection conductors; multiple second connection conductors; and a second semiconductor chip which includes a second internal circuit, multiple second flip-flop circuits connected to the second internal circuit, multiple second selectors, multiple second electrodes respectively connected to the first electrodes via the first connection conductors, and multiple third electrodes connected to respective outputs of the second selectors. The first semiconductor chip further includes multiple fourth electrodes which are respectively connected to the third electrodes via the second connection conductors. At least one of the first semiconductor chip and the second semiconductor chip includes a part of a test circuit unit. The test circuit unit includes a first detection circuit which receives a signal from each of the second electrodes, a first selector control circuit which controls the first selectors, a first expected value generation circuit which generates a first expected value signal including a first expected value, a second detection circuit which receives a signal from each of the fourth electrodes, a second selector control circuit which controls the second selectors, a second expected value generation circuit which generates a second expected value signal including a second expected value, and a test circuit which receives outputs of the first detection circuit and the second detection circuit. Each of the first selectors includes a first signal input which receives a signal from any one of the first flip-flop circuits and a first expected value input which receives the first expected value signal from the first expected value generation circuit. Each of the second selectors includes a

second signal input which receives a signal from any one of the second flip-flop circuits and a second expected value input which receives the second expected value signal from the second expected value generation circuit.
(11) Yet another aspect of the disclosure provides a semiconductor device testing method for a semiconductor device. The semiconductor device includes: a first semiconductor chip which includes a first internal circuit, first flip-flop circuits connected to the first internal circuit, first selectors, and first electrodes connected to respective outputs of the first selectors; first connection conductors; and a second semiconductor chip which includes second electrodes respectively connected to the first electrodes via the first connection conductors and a second internal circuit connected to at least one of the second electrodes. At least one of the first semiconductor chip and the second semiconductor chip includes a part of a test circuit unit. The test circuit unit includes a first detection circuit which receives a signal from each of the second electrodes, a first selector control circuit which controls the first selectors, a first expected value generation circuit which generates a first expected value signal including a first expected value, and a test circuit which receives an output of the first detection circuit. The first expected value generation circuit generates and supplies the first expected value signal of the same first expected value to all of the first selectors, the first selector control circuit controls the first selectors to output the first expected value signal input to a first expected value input of each of the first selectors, the first detection circuit generates a value indicating a first detection result based on a signal input from each of the second electrodes, and the test circuit tests whether or not the first expected value matches a value indicating the first detection result.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** is a diagram showing a semiconductor device according to an embodiment of the disclosure.
(2) FIG. **2**A and FIG. **2**B are diagrams showing specific examples of a logic circuit of the semiconductor device according to the embodiment of the disclosure.
(3) FIG. **3**A and FIG. **3**B are diagrams showing specific examples of the logic circuit of the semiconductor device according to the embodiment of the disclosure.
(4) FIG. **4**A and FIG. **4**B are diagrams showing specific examples of the logic circuit of the semiconductor device according to the embodiment of the disclosure.
(5) FIG. **5** is a diagram showing a semiconductor device not including a first selector.
(6) FIG. **6** is a diagram showing a semiconductor device according to another embodiment of the disclosure.
DESCRIPTION OF THE EMBODIMENTS
(7) According to the disclosure, the semiconductor device and the semiconductor device testing method are capable of shortening the time for testing whether the connection between semiconductor chips is good or bad.
(8) Hereinafter, each embodiment for carrying out the disclosure will be described with reference to the drawings.
(9) FIG. **1** is a diagram showing a semiconductor device according to an embodiment of the disclosure.
(10) A semiconductor device **11** includes a plurality of semiconductor chips and a plurality of first connection conductors **17**. In this embodiment, the semiconductor device includes a first semiconductor chip **13** and a second semiconductor chip **15** as the plurality of semiconductor chips and includes first connection conductors **17***a* to **17***d* as the plurality of first connection conductors **17**. The semiconductor device **11** can be, for example, a three-dimensional (3D) IC or a multi-chip

package (MCP). Further, the semiconductor device may further include a support component **10** such as a package for mounting a plurality of semiconductor chips in three dimensions (3D) or in a multi-chip package. The support component **10** can support the first semiconductor chip **13**, the second semiconductor chip **15**, and the plurality of first connection conductors **17** which are assembled. As will be understood from the subsequent description, the plurality of first connection conductors **17** can include wires, bumps, through wires, and the like for connecting the first semiconductor chip **13** and the second semiconductor chip **15** to each other.

(11) The first semiconductor chip **13** includes a plurality of first flip-flop circuits **19**, a plurality of first selectors **21**, a first internal circuit **23**, and a plurality of first electrodes **25**. The first internal circuit **23** can include a circuit such as a logic circuit configured to provide the main function of the first semiconductor chip **13**. The plurality of first flip-flop circuits **19** is connected to the first internal circuit **23**. In this embodiment, the first flip-flop circuit **19** can be, for example, a D-type flip-flop circuit and the D-type flip-flop circuit includes an input SI, an input D, and an output Q and can include an output Q_ (Q bar) (not shown) if necessary. The input (for example, the input D) of the plurality of first flip-flop circuits **19** is connected to the first internal circuit **23** and the output (for example, the output Q) of the plurality of first flip-flop circuits **19** is connected to each of the first selector circuits **21**. The plurality of first electrodes **25** can include, for example, conductors for external connection such as pad electrodes and bump electrodes. The plurality of first electrodes **25** is connected to each of the first selectors **21**.

(12) The second semiconductor chip **15** includes a plurality of second flip-flop circuits **29**, a second internal circuit **33**, and a plurality of second electrodes **35**. The second internal circuit **33** can be connected to at least one of the plurality of second electrodes **35**. The plurality of second electrodes **35** can include, for example, conductors for external connection such as pad electrodes and bump electrodes.

(13) The first semiconductor chip **13** and the second semiconductor chip **15** are connected to each other via the plurality of first connection conductors **17**. Specifically, the plurality of second electrodes **35** of the second semiconductor chip **15** is respectively connected to the plurality of first electrodes **25** of the first semiconductor chip **13** via the plurality of first connection conductors **17**.

(14) At least one of the first semiconductor chip **13** and the second semiconductor chip **15** includes a test circuit **27**. FIG. **1** shows a case in which the test circuit **27** is provided in the second semiconductor chip **15**.

(15) The semiconductor device **11** can include a first detection circuit **37**, a first selector control circuit **39**, and a first expected value generation circuit **41***a*. The test circuit **27**, the first detection circuit **37**, the first selector control circuit **39**, and the first expected value generation circuit **41***a* are examples of the test circuit unit. The first detection circuit **37** receives signals (S1 to S4) from the plurality of second electrodes **35**. The first selector control circuit **39** controls the plurality of first selectors **21**. The first expected value generation circuit **41***a* generates expected value signals of expected values [0] or [1] to be provided to the plurality of first selectors **21**.

(16) The test circuit unit may be provided in any one of the first semiconductor chip **13** and the second semiconductor chip **15** or may be provided in a part of both of the first semiconductor chip **13** and the second semiconductor chip **15**. That is, at least one of the first semiconductor chip **13** and the second semiconductor chip **15** includes a part of the test circuit unit. FIG. **1** shows a case in which the first selector control circuit **39** and the first expected value generation circuit **41***a* are provided in the first semiconductor chip **13** as a part of the test circuit unit and the test circuit **27** and the first detection circuit **37** are provided in the second semiconductor chip **15** as a part of the test circuit unit.

(17) Each of the plurality of first selectors **21** includes a first signal input **21***a*, a first expected value input **21***b*, a first selector output **21***c*, and a first selection input **21***d*. The first signal input **21***a* is connected to any one output (for example, the output Q) of the plurality of first flip-flop circuits **19** and receives a signal from the output. The first expected value input **21***b* receives an expected value

signal of an expected value [0] or [1] from the first expected value generation circuit **41***a*. The first selector output **21***c* is connected to the first electrode **25**. Each of the plurality of first selectors **21** connects the first expected value input **21***b* to the first selector output **21***c* in a test mode according to the embodiment that uses the test circuit **27**. The first selector **21** connects the first signal input **21***a* to the first selector output **21***c* in an operation mode different from the test mode.
(18) According to the semiconductor device **11**, the first selector control circuit **39** controls the plurality of first selectors **21** in the test mode so that the plurality of first selectors **21** pass the expected value signal from the first expected value generation circuit **41***a*. The expected value signal passed through the first selector **21** is provided to the first electrode **25**. The first detection circuit **37** receives each of the signals (S1 to S4) from the second electrode **35**. When the expected value signal from the first expected value generation circuit **41***a* to the first expected value input **21***b* can be given to the first detection circuit **37** via the plurality of first electrodes **25** of the first semiconductor chip **13** and the plurality of second electrodes **35** of the second semiconductor chip **15**, it is possible to test whether or not the first semiconductor chip **13** and the second semiconductor chip **15** are properly connected via the plurality of first connection conductors **17**.
(19) In the semiconductor device **11**, for example, each of the plurality of second electrodes **35** is configured to receive an expected value signal of an expected value [**0**] and [**1**] from the first selector **21** via the first electrode **25** and the first connection conductor **17**. In the semiconductor device **11** including interconnections by four first connection conductors **17** between four first electrodes **25** and four second electrodes **35**, the first detection circuit **37** receives four signals (S1 to S4) from four second electrodes **35**. The first detection circuit **37** can detect the value of the signal input based on four signals (S1 to S4) from four second electrodes **35** and provide a value indicating the detection result.
(20) The first detection circuit **37** can include, for example, at least one of a first OR circuit **43** and a first AND circuit **45**.
(21) The first OR circuit **43** receives four signals (S1 to S4) from four second electrodes **35** and generates an OR result of four signals (S1 to S4) from four second electrodes **35**. The first OR circuit **43** generates a value [0] indicating the detection result in response to the expected value signals of all expected values [0] from the plurality of first selectors **21** and generates and outputs a value [1] indicating the detection result in response to the expected value signal of at least one expected value [1] from the plurality of first selectors **21**. That is, a value “1” indicating the detection result is output as the output value of the first OR circuit **43**. A value [0] or [1] indicating these detection results is provided to the test circuit **27** and the test circuit **27** tests whether or not the output value of the first OR circuit **43** matches the expected value for the input expected value signal.
(22) The first AND circuit **45** receives four signals (S1 to S4) from four second electrodes **35** and generates an AND result of four signals (S1 to S4) from four second electrodes **35**. The first AND circuit **45** generates a value [1] indicating the detection result in response to the expected value signals of all expected values [1] from the plurality of first selectors **21** and generates and outputs a value [0] indicating the detection result in response to the expected value signal of at least one expected value [0] from the plurality of first selectors **21**. That is, a value [0] indicating the detection result is output as the output value of the first AND circuit **45**. A value [0] or [1] indicating these detection results is provided to the test circuit **27** and the test circuit **27** tests whether or not the output value of the first AND circuit **45** matches the expected value for the input expected value signal.
(23) FIG. **2**A and FIG. **2**B are diagrams showing the first OR circuit **43** and the plurality of first connection conductors **17** (**17***a*, **17***b*, **17***c*, **17***d*) between the first semiconductor chip **13** and the second semiconductor chip **15** in the semiconductor device **11** according to this embodiment. FIG. **3**A and FIG. **3**B are diagrams showing the first AND circuit **45** and the plurality of first connection conductors **17** (**17***a*, **17***b*, **17***c*, **17***d*) between the first semiconductor chip **13** and the second

semiconductor chip **15** in the semiconductor device **11** according to this embodiment. Referring to FIG. **2**A and FIG. **2**B and FIG. **3**A and FIG. **3**B, the operation of the first OR circuit **43** and the first AND circuit **45** will be described. The first selector control circuit **39** controls the plurality of first selectors **21** so that all of the plurality of first selectors **21** passes the expected value signal from the first expected value generation circuit **41***a*.
(24) Referring to FIG. **2**A and FIG. **2**B, four first electrodes **25** and four second electrodes **35** are connected to each other by four first connection conductors **17***a*, **17***b*, **17***c*, and **17***d* and the signals from four second electrodes **35** are given to the first OR circuit **43**.
(25) Regarding the expected value signals of the expected values [0] or [1] from four first selectors **21** to four first electrodes **25**, when a signal value [1] is always transmitted to the second electrode **35** in at least one of four first connection conductors **17***a*, **17***b*, **17***c*, and **17***d*, for example, the first connection conductor **17***c*, the first OR circuit **43** responses as below.
(26) As shown in FIG. **2**A, the first OR circuit **43** generates a value [1] indicating the detection result in response to the expected value signals of all expected values [0] from four first selectors **21**. Then, a value indicating this detection result is output as the output value of the first OR circuit **43**. A value [1] indicating this detection result is provided to the test circuit **27** and the test circuit **27** generates a test result indicating that the output value of the first OR circuit **43** does not match the expected value [0] for all input expected value signals.
(27) Further, as shown in FIG. **2**B, the first OR circuit **43** generates a value [1] indicating the detection result in response to the expected value signals of all expected values [1] from four first selectors **21**. Then, a value indicating this detection result is output as the output value of the first OR circuit **43**. When a value [1] indicating this detection result is provided to the test circuit **27**, the test circuit **27** generates a test result indicating that the output value of the first OR circuit **43** matches the input expected value [1] for all input expected value signals.
(28) Referring to FIG. **3**A and FIG. **3**B, four first electrodes **25** and four second electrodes **35** are connected to each other by four first connection conductors **17***a*, **17***b*, **17***c*, and **17***d* and the signals from four second electrodes **35** are given to the first AND circuit **45**.
(29) Regarding the expected value signals of the expected values [0] or [1] from four first selectors **21** to four first electrodes **25**, when a signal value [0] is always transmitted to the second electrode **35** in at least one of four first connection conductors **17***a*, **17***b*, **17***c*, and **17***d*, for example, the first connection conductor **17***c*, the first AND circuit **45** responses as below.
(30) As shown in FIG. **3**A, the first AND circuit **45** generates a value [0] indicating the detection result in response to the expected value signals of all expected values [1] from four first selectors **21**. A value [0] indicating this detection result is provided to the test circuit **27** and the test circuit **27** generates a test result indicating that the output value of the first AND circuit **45** does not match the expected value [1] for all input expected value signals.
(31) Further, as shown in FIG. **3**B, the first AND circuit **45** generates a value [0] indicating the detection result in response to the expected value signals of all expected values [0] from four first selectors **21**. Then, a value indicating this detection result is output as the output value of the first AND circuit **45**. A value [0] indicating this detection result is provided to the test circuit **27** and the test circuit **27** generates a test result indicating that the output value of the first AND circuit **45** matches the expected value [0] for all input expected value signals.
(32) It has been described that an interconnection always transmitting a signal value [0] to the second electrode **35** and an interconnection always transmitting a signal value [1] thereto can be detected while describing the operation of the 4-input OR circuit and the 4-input AND circuit. The semiconductor device **11** can use an N-input OR circuit and an N-input AND circuit instead of the 4-input OR circuit **43** and the 4-input AND circuit **45**.
(33) When an expected value signal of an expected value [0] is given to all of the first expected value inputs **21***b* of N number of the first selectors **21** and the expected value signal of this expected value is transmitted from the first electrode **25** to the second electrode **35**, the first OR

circuit **43** generates a signal with a logic value of "0". However, when the expected value signal of this expected value is not transmitted from the first electrode **25** to the second electrode **35**, specifically, at least one of N number of interconnections transmits a signal value [1] from the first electrode to the second electrode **35**, a signal with a logic value of "1" is generated. The first OR circuit **43** can be used to detect that at least one of N number of interconnections always gives a signal value [1] to the second electrode **35**.
(34) When an expected value signal of an expected value [1] is given to all of the first expected value inputs **21***b* of N number of the first selectors **21** and the expected value signal of this expected value is transmitted from the first electrode **25** to the second electrode **35**, the first AND circuit **45** generates a signal with a logic value of "1". However, when the expected value signal of this expected value is not transmitted from the first electrode **25** to the second electrode **35**, specifically, at least one of N number of interconnections transmits a signal value [0] from the first electrode to the second electrode **35**, a signal with a logic value of "0" is generated. The first AND circuit **45** can be used to detect that at least one of N number of interconnections always gives a signal value [0] to the second electrode **35**.
(35) It has been described that an interconnection always transmitting a signal value [0] to the second electrode **35** and an interconnection always transmitting a signal value [1] thereto are detected by using the plurality of first selectors **21** and the first detection circuit **37**.
(36) Next, a case of detecting a connection position of an interconnection always transmitting a signal value [0] and an interconnection always transmitting a signal value [1] will be described.
(37) As shown in FIG. **1**, the first detection circuit **37** can include a first logic circuit **47** in addition to the first OR circuit **43** and the first AND circuit **45**. The first logic circuit **47** receives four signals (S1 to S4) from four second electrodes **35** and generates a set of values based on four signals (S1 to S4) from four second electrodes **35**. The first logic circuit **47** generates and outputs a value indicating the detection result in response to the expected value signals of the expected values [0] or [1] from the plurality of first selectors **21**. That is, a value indicating the detection result is output as the output value of the first logic circuit **47**. A value indicating the detection result is provided to the test circuit **27** and the test circuit **27** tests whether or not a set of output values of the first logic circuit **47** match a set of expected values for the input expected value signals. The first expected value generation circuit **41***a* provides expected value signals of independent expected values [0] or [1] to the first expected value inputs **21***b* of the plurality of first selectors **21**.
(38) Again, a description will be made by exemplifying a case in which the expected value signals of the expected values [0] or [1] from four first selectors **21** to four first electrodes **25** are always transmitted to the second electrode **35** as a signal value [0] in at least one of four first connection conductors **17***a*, **17***b*, **17***c*, and **17***d*, for example, the first connection conductor **17***c*.
(39) FIG. **4**A and FIG. **4**B are diagrams showing the first logic circuit **47** and the plurality of first connection conductors **17** (**17***a*, **17***b*, **17***c*, **17***d*) of the first semiconductor chip **13** and the second semiconductor chip **15** in the semiconductor device **11** according to this embodiment. Referring to FIG. **4**A and FIG. **4**B, the operation of the first logic circuit **47** will be described. The first selector control circuit **39** controls the plurality of first selectors **21** so that all of the plurality of first selectors **21** pass the expected value signal from the first expected value generation circuit **41***a*.
(40) Referring to FIG. **4**A and FIG. **4**B, four first electrodes **25** and four second electrodes **35** are connected to each other by four first connection conductors **17***a*, **17***b*, **17***c*, and **17***d* and signals are given from four second electrodes **35** to the first logic circuit **47**. The first expected value generation circuit **41***a* provides expected value signals of independent expected values to the first expected value inputs **21***b* of four first selectors **21**. When the expected value signal of such an expected value is given, the test circuit **27** tests the response of the first logic circuit **47** to the signal from the second electrode **35**.
(41) As shown in FIG. **4**A, among many processes of providing the expected value signals of various independent expected values from the first expected value generation circuit **41***a*, for

example, the expected value signal of the expected value below is given to each of the first expected value inputs **21***b* of four first selectors **21**. Expected value of first expected value input **21***b* of first selector **21** related to first connection conductor **17***a: [*0] Expected value of first expected value input **21***b* of first selector **21** related to first connection conductor **17***b: [*0] Expected value of first expected value input **21***b* of first selector **21** related to first connection conductor **17***c: [*1] Expected value of first expected value input **21***b* of first selector **21** related to first connection conductor **17***d: [*0]
(42) For the expected value signal of such an expected value, the values of the signals (S1 to S4 according to the reference numerals shown in FIG. **1**) on the second electrode **35** input to the first logic circuit **47** are as below. Value of signal on second electrode **35** related to first connection conductor **17***a: [*0] Value of signal on second electrode **35** related to first connection conductor **17***b: [*0] Value of signal on second electrode **35** related to first connection conductor **17***c: [*0] Value of signal on second electrode **35** related to first connection conductor **17***d: [*0]
(43) The set of the expected values of the expected value signals input to the first expected value inputs **21***b* of four first selectors **21** is [0], [0], [1], and [0] in order in the first connection conductors **17***a* to **17***d* and the set of the values of the signals on four second electrodes **35** is [0], [0], [0], and [0] in order in the first connection conductors **17***a* to **17***d*. While the expected value on the first electrode **25** related to the first connection conductor **17***c* is [1], the value of the signal on the second electrode **35** related to the first connection conductor **17***c* is [0]. The first logic circuit **47** generates a value indicating the detection result based on each of the signals from four second electrodes. Then, a value indicating this detection result is output as the output value of the first logic circuit **47**. A value indicating this detection result is provided to the test circuit **27** and the test circuit **27** generates a test result indicating that the set of the output values of the first logic circuit **47** does not match the set of the input expected values.
(44) Further, as shown in FIG. **4**B, among many processes of providing the expected value signals of various independent expected values from the first expected value generation circuit **41***a*, for example, the expected value signal of the expected value below is given to each of the first expected value inputs **21***b* of four first selectors **21**. Expected value of first expected value input **21***b* of first selector **21** related to first connection conductor **17***a: [*1] Expected value of first expected value input **21***b* of first selector **21** related to first connection conductor **17***b: [*1] Expected value of first expected value input **21***b* of first selector **21** related to first connection conductor **17***c: [*0] Expected value of first expected value input **21***b* of first selector **21** related to first connection conductor **17***d: [*1]
(45) For the expected value signal of such an expected value, the values of the signals (S1 to S4 according to the reference numerals shown in FIG. **1**) on the second electrode **35** input to the first logic circuit **47** are as below. Value of signal on second electrode **35** related to first connection conductor **17***a: [*1] Value of signal on second electrode **35** related to first connection conductor **17***b: [*1] Value of signal on second electrode **35** related to first connection conductor **17***c: [*0] Value of signal on second electrode **35** related to first connection conductor **17***d: [*1]
(46) The set of the expected values of the expected value signals input to the first expected value inputs **21***b* of four first selectors **21** is [1], [1], [0], and [1] in order in the first connection conductors **17***a* to **17***d* and the set of the values of the signals on four second electrodes **35** is [1], [1], [0], and [1] in order in the first connection conductors **17***a* to **17***d*. While the expected value on the first electrode **25** related to the first connection conductor **17***c* is [0], the value of the signal on the second electrode **35** related to the first connection conductor **17***c* is also [0]. The first logic circuit **47** generates a value indicating the detection result based on each of the signals from four second electrodes. Then, a value indicating this detection result is output as the output value of the first logic circuit **47**. A value indicating this detection result is provided to the test circuit **27** and the test circuit **27** generates a test result indicating that the set of the output values of the first logic circuit **47** matches the set of the input expected values.

(47) Further, when the expected value signals of the expected values [0] or [1] from four first selectors **21** to four first electrodes **25** are always transmitted to the second electrode **35** as a signal value [1] in at least one of four first connection conductors **17***a*, **17***b*, **17***c*, and **17***d*, for example, the first connection conductor **17***c*, the description can be said in an opposite manner. That is, the value of the signal on the first electrode **25** related to the first connection conductor **17***c* is [1] while the expected value on the first electrode **25** related to the first connection conductor **17***c* is an expected value [1] and the test circuit **27** generates a test result indicating that the set of the output values of the first logic circuit **47** matches the set of the input expected values. Further, the value of the signal on the first electrode **25** related to the first connection conductor **17***c* is an expected value [1] while the expected value on the first electrode **25** related to the first connection conductor **17***c* is [0] and the test circuit **27** generates a test result indicting that the set of the output values of the first logic circuit **47** does not match the set of the input expected values.
(48) The first expected value generation circuit **41***a* supplies a first value [0] or [1] to one of four first selectors **21** as the expected value and supplies a second value [1] or [0] different from the first value [0] or [1] to the other three first selectors **21** as the expected values. The first logic circuit **47** receives signals from four second electrodes **35** and generates a value indicating the detection result in response to the supplied signals. A value indicating this detection result is provided to the test circuit **27** and the test circuit **27** tests whether or not the set of the values indicating the detection result corresponding to each of four first selectors **21** matches the set of the input expected values. It is possible to identify the connection conductor which does not match the expected value by sequentially applying a combination of various sets of the expected values to the first expected value input **21***b* of the first selector **21**.
(49) It has been described that the connection positions of an interconnection always transmitting a signal value [0] to the second electrode **35** and an interconnection always transmitting a signal value [1] thereto can be detected while describing the operation of the 4-input OR circuit. The semiconductor device **11** can use an N-input logic circuit instead of the 4-input logic circuit.
(50) When an expected value signal of an expected value [1] is given to one of the first expected value inputs **21***b* of N number of the first selectors **21**, an expected value signal of an expected value [0] is given to the other three first selectors **21**, and the expected value signal of this expected value is transmitted from the first electrode **25** to the second electrode **35**, the first logic circuit **47** generates a set of signals of values corresponding to a set of the expected values input based on the signals respectively given to N number of the first selectors **21**. However, when the expected value signal of this expected value is not transmitted from the first electrode **25** to the second electrode **35**, specifically, an interconnection corresponding to the first selector **21** receiving an expected value signal of an expected value [1] transmits a signal value [0] from the first electrode **25** to the second electrode **35**, a set of signals of values not corresponding to the set of input expected values is generated based on the signals respectively given to N number of the first selectors **21**. The first logic circuit **47** can be used to detect that an interconnection corresponding to the first selector receiving an expected value signal of an expected value [1] always gives a signal value [0].
(51) When an expected value signal of an expected value [0] is given to one of the first expected value inputs **21***b* of N number of the first selectors **21**, an expected value signal of an expected value [1] is given to the other three first selectors **21**, and the expected value signal of this expected value is transmitted from the first electrode **25** to the second electrode **35**, the first logic circuit **47** generates a set of signals of values corresponding to a set of the expected values input based on the signals respectively given to N number of the first selectors **21**. However, when the expected value signal of the expected value is not transmitted from the first electrode **25** to the second electrode **35**, specifically, an interconnection corresponding to the first selector **21** receiving an expected value signal of an expected value [0] transmits a signal value [1] from the first electrode **25** to the second electrode **35**, a set of signals of values not corresponding to the set of input expected values is generated based on the signals respectively given to N number of the first selectors **21**. The first

logic circuit **47** can be used to detect that an interconnection corresponding to the first selector receiving an expected value signal of an expected value [0] always gives a signal value [1].
(52) A first encoder circuit **49** that generates a signal value corresponding to the number of the first selectors **21** connected to the first electrodes **25** related to all interconnections to be inspected may be provided to the first expected value generation circuit **41***a* in order to generate such a combination of expected values. Further, a decoder circuit that generates a value based on the signals from the plurality of second electrodes **35** by receiving the signals from the second electrodes **35** related to all interconnections to be inspected may be provided to the first logic circuit **47** in order to analyze the mismatch of the values of the generated signals. The signal from this decoder circuit is provided to the test circuit **27** in order to test the value of the signal from the second electrode **35**.
(53) Again, the semiconductor device **11** will be described with reference to FIG. **1**. In the second semiconductor chip **15**, the second internal circuit **33** may include a circuit, for example, a logic circuit configured to provide a main function of the second semiconductor chip **15**. The second flip-flop circuit **29** is connected to the second internal circuit **33**. In this embodiment, the second flip-flop circuit **29** can be, for example, a D-type flip-flop circuit and this D-type flip-flop circuit includes an input SI, an input D, and an output Q and can include an output Q_ (Q bar) output (not shown) if necessary. The signals (S1 to S4) of the plurality of second electrodes **35** are connected to the inputs (for example, the inputs SI) of the second flip-flop circuits **29** and are connected to the second internal circuit **33**. The input (for example, the input D) of the second flip-flop circuit **29** is connected to the second internal circuit.
(54) The outputs (for example, the outputs Q) of the plurality of second flip-flop circuits **29** are connected to the inputs (for example, the inputs SI) of the plurality of first flip-flop circuits **19** via the plurality of third electrodes **34** of the second semiconductor chip **15**, the plurality of second connection conductors **16**, and the plurality of fourth electrodes **24** of the first semiconductor chip **13** and are connected to the first internal circuit **23**.
(55) Further, the SIN line from the test circuit **27** is connected to any one input (for example, the input SI) of the plurality of first flip-flop circuits **19** via the third electrode **34** of the second semiconductor chip **15**, the second connection conductor **16**, and the fourth electrode **24** of the first semiconductor chip **13**.
(56) The test circuit **27** receives an output value OR2 of the first OR circuit **43**, an output value AND2 of the first AND circuit **45**, and a signal SOUT from at least one output (for example, the output Q) of the plurality of first flip-flop circuits **19** via the selector **20**. The selector **20** selects a signal to be provided to the test circuit **27** and to be tested from the output value OR2 of the first OR circuit **43**, the output value AND2 of the first AND circuit **45**, and the signal SOUT from the first flip-flop circuit **19** according to the content of the test.
(57) Further, the test circuit **27** receives the output of the first logic circuit **47**, specifically, the signal from the decoder circuit via an expected value bus **52***a* (decoded signal).
(58) The test circuit **27** can receive signals related to the test pattern, the test mode, and the expected value via the interface IF with the outside of the semiconductor device **11** or can output the signals.
(59) The test circuit **27** can be connected to the selector control circuit **39** via a fifth electrode **36** of the second semiconductor chip **15**, a third connection conductor **18**, and a sixth electrode **26** of the first semiconductor chip **13**.
(60) The test circuit **27** is connected to the first expected value generation circuit **41***a* (for example, the encoder circuit) via an input value bus **52***b* that passes through the fifth electrode **36** of the second semiconductor chip **15**, the third connection conductor **18**, and the sixth electrode **26** of the first semiconductor chip **13**. The first expected value generation circuit **41***a* (specifically, the encoder circuit) generates N number of the expected value signals including the expected values from the signals on the input value bus **52***b*.

(61) The electrodes **24**, **26**, **34**, and **36** can include conductors for external connection such as pad electrodes and bump electrodes similarly to the above-described electrodes (**25**, **35**). In the semiconductor device **11**, the test circuit **27** provides signals on the input line bus **52***b* to the first expected value generation circuit **41***a* and tests the results from the expected value bus **52***a* in order to generate independent expected values. Specifically, the signals (S1 to S4) input from the plurality of second electrodes **35** of the first semiconductor chip **13** are decoded and the decoded signals are given from the expected value bus **52***a* to the test circuit **27** and are evaluated in the test circuit **27**.
(62) Referring to FIG. **1**, for example, a combinational logic circuit **50** is disposed on each of the signal lines related to the first connection conductors **17***b* and **17***c*. The combinational logic circuit **50** receives a signal from the output (for example, the output Q) of the first flip-flop circuit **19** and is connected to the first signal input **21***a* of the first selector **21**.
(63) The semiconductor device **11** interrupts signals from the first flip-flop circuit **19** and the combinational logic circuit **50** by using the first selector **21** connected to the first connection conductor **17**. Hence, the mismatch of the expected values related to the first connection conductor **17** is not caused by the combinational logic circuit **50** and/or the first flip-flop circuit **19**. For example, it is possible to determine whether or not there is a problem with the combinational logic circuit **50** from a test related to a scan chain.
(64) FIG. **5** is a diagram showing a semiconductor device not provided with a plurality of first selectors. In the subsequent description, for ease of understanding, the reference numerals used in the semiconductor device **11** of FIG. **1** are used when it is possible to use the reference numerals. Referring to FIG. **5**, a semiconductor device **51** is shown. The semiconductor device **51** includes a semiconductor chip **12** and a semiconductor chip **14** instead of the first semiconductor chip **13** and the second semiconductor chip **15**. Each of the semiconductor chip **12** and the semiconductor chip **14** is not provided with the plurality of first selectors **21** of the semiconductor device **11**. For example, the output (for example, the output Q) of the first flip-flop circuit **19** is directly connected to the first electrode **25**. Further, the combinational logic circuit **50** is directly connected to the first electrode **25**.
(65) In the semiconductor device **51**, the mismatch of the expected value related to the first connection conductor **17** is tested when the test pattern provided as the SIN signal from the test circuit **27** is returned to the test circuit **27** as the signal SOUT via the plurality of first flip-flop circuits (**19**, **29**) in the test related to the scan chain. Hence, the test becomes clear after propagating through one or a plurality of flip-flop circuit or circuits located on the downstream side of the first connection conductor **17** that caused the mismatch of the expected values. Hence, the transfer time of the stage number of the downstream flip-flop circuit is required until the determination. Further, when there is the combinational logic circuit **50** between the flip-flop circuits (**19**, **29**) on the propagation line of the test pattern, it cannot be determined which of the first connection conductor **17** and the combinational logic circuit **50** causes the mismatch of the expected value.
(66) According to the semiconductor device **11** of FIG. **1**, since the mismatch of the expected value related to the first connection conductor **17** is determined by inputting an expected value signal including an expected value to the selector circuit **21** in the test mode, it is possible to determine the mismatch of the expected value related to the first connection conductor **17** without delays related to the propagation of flip-flop circuits and regardless of the presence or absence of failures in the combinational logic circuit **50**.
(67) FIG. **6** is a diagram showing a semiconductor device according to another embodiment of the disclosure. In the semiconductor device **11** shown in FIG. **1**, the plurality of first selectors **21** is arranged on the signal path for sending signals from the first semiconductor chip **13** to the second semiconductor chip **15**. Referring to FIG. **6**, in a semiconductor device **11***a*, a plurality of second selectors **22** is arranged on a signal path for sending signals from the second semiconductor chip **15** to the first semiconductor chip **13** in addition to the plurality of first selectors **21** of the

semiconductor device **11** shown in FIG. **1**. Hence, the plurality of second selectors **22** is provided in the second semiconductor chip **15**.
(68) The semiconductor device **11***a* can include a second detection circuit **38**, a second selector control circuit **40**, and a second expected value generation circuit **41***b* similarly to the semiconductor device **11**. The second detection circuit **38**, the second selector control circuit **40**, and the second expected value generation circuit **41***b* are examples of the test circuit unit together with the test circuit **27**, the first detection circuit **37**, the first selector control circuit **39**, and the first expected value generation circuit **41***a*.
(69) In this embodiment, additional circuits for testing the expected values, for example, the second detection circuit **38**, the second selector control circuit **40**, and the second expected value generation circuit **41***b* are provided to the second semiconductor chip **15** while adding the plurality of second selectors **22**. The second selector control circuit **40** controls the plurality of second selectors **22**. The second expected value generation circuit **41***b* generates expected value signals of expected values [0] or [1] provided to the second selector **22**.
(70) Each of the plurality of second selectors **22** includes a second signal input **22***a*, a second expected value input **22***b*, a second selector output **22***c*, and a second selection input **22***d*. The second signal input **22***a* is connected to the output (for example, the output Q) of any one of the plurality of second flip-flop circuits **29** and receives a signal from the output. The second expected value input **22***b* receives an expected value signal of expected value [0] or [1] from the second expected value generation circuit **41***b*. The second selector output **22***c* is connected to the third electrode **34**. The plurality of second selectors **22** connect the second expected value input **22***b* to the second selector output **22***c* in the test mode according to the embodiment that uses the test circuit **27**. The plurality of second selectors **22** connect the second signal input **22***a* to the second selector output **22***c* in an operation mode different from the test mode.
(71) Each of the plurality of third electrodes **34** receives a signal from the output of the second selector **22** and is connected to the fourth electrode **24** via the second connection conductor **16**. The second detection circuit **38** receives each of the signals (T1 to T4) from the fourth electrodes **24** on the first semiconductor chip **13**.
(72) According to the semiconductor device **11***a*, the second selector control circuit **40** controls the plurality of second selectors **22** so that the plurality of second selectors **22** pass the expected value signal from the second expected value generation circuit **41***b* in the test mode. The expected value signal having passed through the second selector **22** is provided to the third electrode **34**. The second detection circuit **38** receives each of the signals (T1 to T4) from the fourth electrode **24**. When the expected value signal from the second expected value generation circuit **41***b* to the second expected value input **22***b* is given to the second detection circuit **38** via the plurality of third electrodes **34** of the second semiconductor chip **15** and the plurality of fourth electrodes **24** of the first semiconductor chip **13**, it is possible to test whether or not the first semiconductor chip **13** and the second semiconductor chip **15** are properly connected via the second connection conductor **16**.
(73) In the semiconductor device **11***a* including the interconnection by four second connection conductors **16** between four third electrodes **34** and four fourth electrodes **24**, the second detection circuit **38** receives four signals (T1 to T4) from four fourth electrodes **24**. In the semiconductor device **11***a*, for example, when the fourth electrode **24** is configured to receive an expected value signal of an expected value [0] and [1] from the second selector **22** via the third electrode **34** and the second connection conductor **16**, the second detection circuit **38** can detect the value of the signal input based on four signals (S1 to S4) from four fourth electrodes **24** and provide a value indicating the detection result.
(74) The second detection circuit **38** can include, for example, at least one of the second OR circuit **42** and the second AND circuit **44**. The second detection circuit **38** enables the same detection as the first detection circuit **37** for the plurality of second connection conductors **16**. The second OR circuit **42** receives four signals (T1 to T4) from four fourth electrodes **24** and generates an OR

result of four signals (T1 to T4) from four fourth electrodes **24**. Further, the second AND circuit **44** receives four signals (T1 to T4) from four fourth electrodes **24** and generates an AND result of four signals (T1 to T4) from four fourth electrodes **24**.
(75) In this embodiment, the output value OR1 of the second OR circuit **42** and the output value AND1 of the second AND circuit **44** are connected to the test circuit **27** via the selector **28**, the sixth electrode **26**, the third connection conductor **18**, and the fifth electrode **36**. The selector **28** selects the signals from the second OR circuit **42** and the second AND circuit **44** and provides the signals to the test circuit **27**.
(76) According to such a connection, the test described with reference to FIG. **2**A to FIG. **3**B is enabled for the second connection conductor **16** using the second OR circuit **42** and the second AND circuit **44**.
(77) The second detection circuit **38** can include a second logic circuit **46** similarly to the semiconductor device **11**. The second logic circuit **46** receives four signals (T1 to T4) from four fourth electrodes **24** and generates a set of values based on four signals (T1 to T4) from four fourth electrodes **24**. The second logic circuit **46** can have the same configuration as the first logic circuit **47**. In this embodiment, the second expected value generation circuit **41***b* is configured to generate a combination of sets of expected values in a manner similar to the second expected value generation circuit **41***a*. The second selector control circuit **40** is configured to control the plurality of second selectors **22** in a manner similar to the first selector control circuit **39**.
(78) According to such a configuration, the test described with reference to FIG. **4**A and FIG. **4**B is enabled for the second connection conductor **16** using the second logic circuit **46**.
(79) A second encoder circuit **48** that generates a signal value corresponding to the number of the second selectors **22** connected to the third electrode **34** related to all interconnections to be inspected may be provided to the second expected value generation circuit **41***b* in order to generate such a combination of expected values. Further, a decoder circuit that generates a value based on the signals from the plurality of fourth electrodes by receiving the signals from the fourth electrodes **24** related to all interconnections to be inspected may be provided to the second logic circuit **46** in order to analyze the mismatch of the expected values to be generated. In this embodiment, the signal from this decoder circuit is provided to the test circuit **27** via the sixth electrode **26**, the third connection conductor **18**, and the fifth electrode **36** in order to test the value of the signal from the fourth electrode **24**.
(80) If necessary, the combinational logic circuit **50** may be connected on the path that connects the second signal input **22***a* of the second selector **22** and the output (for example, the output Q) of the second flip-flop **29**.
(81) According to the semiconductor device **11***a* of FIG. **6**, since the mismatch of the expected values related to the first connection conductors **17** and **16** is determined by inputting an expected value signal including an expected value to the selector circuits **21** and **22** in the test mode similarly to the semiconductor device **11** of FIG. **1**, it is possible to determine the mismatch of the expected values related to the first connection conductors **17** and **16** without delays related to the propagation of flip-flop circuits and regardless of the presence or absence of failures in combinational logic circuits.
(82) As described above, according to this embodiment, there is provided the semiconductor device capable of shortening the time for testing whether the connection between the semiconductor chips is good or bad.
(83) In FIGS. **1** and **6**, the first detection circuit **37**, the first selector control circuit **39**, the first expected value generation circuit **41***a*, the second detection circuit **38**, the second selector control circuit **40**, and the second expected value generation circuit **41***b* are drawn outside the test circuit **27**, but the test circuit **27** can include the first detection circuit **37**, the first selector control circuit **39**, the first expected value generation circuit **41***a*, the second detection circuit **38**, the second selector control circuit **40**, and the second expected value generation circuit **41***b*. In this case, the

test circuit **27** can be provided in at least one of the first semiconductor chip **13** and the second semiconductor chip **15**.

(84) It will be apparent to those skilled in the art that various modifications and variations can be made to the disclosed embodiments without departing from the scope or spirit of the disclosure. In view of the foregoing, it is intended that the disclosure covers modifications and variations provided that they fall within the scope of the following claims and their equivalents.

Claims

1. A semiconductor device comprising: a first semiconductor chip which includes a first internal circuit, a plurality of first flip-flop circuits connected to the first internal circuit, a plurality of first selectors, and a plurality of first electrodes connected to respective outputs of the plurality of first selectors; a plurality of first connection conductors; and a second semiconductor chip which includes a plurality of second electrodes respectively connected to the plurality of first electrodes via the plurality of first connection conductors and a second internal circuit connected to at least one of the plurality of second electrodes, wherein at least one of the first semiconductor chip and the second semiconductor chip includes a part of a test circuit unit, wherein the test circuit unit includes a first detection circuit which receives a signal from each of the plurality of second electrodes, a first selector control circuit which controls the plurality of first selectors, a first expected value generation circuit which generates a first expected value signal including a first expected value, and a test circuit which receives an output of the first detection circuit, and wherein each of the plurality of first selectors includes a first signal input which receives a signal from any one of the plurality of first flip-flop circuits and a first expected value input which receives the first expected value signal from the first expected value generation circuit.

2. The semiconductor device according to claim 1, wherein each of the plurality of first selectors outputs the first expected value signal input to the first expected value input in a test mode and outputs a signal which is from any one of the plurality of first flip-flop circuits and input to the first signal input in an operation mode different from the test mode.

3. The semiconductor device according to claim 1, wherein the first detection circuit includes a first OR circuit which receives the signal from each of the plurality of second electrodes and generates an OR result of the signals from the plurality of second electrodes.

4. The semiconductor device according to claim 1, wherein the first detection circuit includes a first AND circuit which receives the signal from each of the plurality of second electrodes and generates an AND result of the signals from the plurality of second electrodes.

5. The semiconductor device according to claim 1, wherein the first expected value generation circuit includes a first encoder circuit which generates a signal value corresponding to the number of the plurality of first selectors, and wherein the first detection circuit includes a first decoder circuit which receives the signals from the plurality of second electrodes and generates a value based on the signals from the plurality of second electrodes.

6. The semiconductor device according to claim 1, wherein the first semiconductor chip further includes at least one combinational logic circuit, and wherein in at least one first selector of the plurality of first selectors, a signal from at least one first flip-flop circuit of the plurality of first flip-flop circuits is input via the combinational logic circuit of the first semiconductor chip.

7. A semiconductor device comprising: a first semiconductor chip which includes a first internal circuit, a plurality of first flip-flop circuits connected to the first internal circuit, a plurality of first selectors, and a plurality of first electrodes connected to respective outputs of the plurality of first selectors; a plurality of first connection conductors; a plurality of second connection conductors; and a second semiconductor chip which includes a second internal circuit, a plurality of second flip-flop circuits connected to the second internal circuit, a plurality of second selectors, a plurality of second electrodes respectively connected to the plurality of first electrodes via the plurality of

first connection conductors, and a plurality of third electrodes connected to respective outputs of the plurality of second selectors, wherein the first semiconductor chip further includes a plurality of fourth electrodes which are respectively connected to the plurality of third electrodes via the plurality of second connection conductors, wherein at least one of the first semiconductor chip and the second semiconductor chip includes a part of a test circuit unit, wherein the test circuit unit includes a first detection circuit which receives a signal from each of the plurality of second electrodes, a first selector control circuit which controls the plurality of first selectors, a first expected value generation circuit which generates a first expected value signal including a first expected value, a second detection circuit which receives a signal from each of the plurality of fourth electrodes, a second selector control circuit which controls the plurality of second selectors, a second expected value generation circuit which generates a second expected value signal including a second expected value, and a test circuit which receives outputs of the first detection circuit and the second detection circuit, wherein each of the plurality of first selectors includes a first signal input which receives a signal from any one of the plurality of first flip-flop circuits and a first expected value input which receives the first expected value signal from the first expected value generation circuit, and wherein each of the plurality of second selectors includes a second signal input which receives a signal from any one of the plurality of second flip-flop circuits and a second expected value input which receives the second expected value signal from the second expected value generation circuit.
8. The semiconductor device according to claim 7, wherein each of the plurality of second selectors outputs the second expected value signal input to the second expected value input in a test mode and outputs a signal that is from any one of the plurality of second flip-flop circuits and input to the second signal input in an operation mode different from the test mode.
9. The semiconductor device according to claim 7, wherein the second detection circuit includes a second OR circuit which receives a signal from each of the plurality of fourth electrodes and generates an OR result of the signals from the plurality of fourth electrodes.
10. The semiconductor device according to claim 7, wherein the second detection circuit includes a second AND circuit which receives a signal from each of the plurality of fourth electrodes and generates an AND result of the signals from the plurality of fourth electrodes.
11. The semiconductor device according to claim 7, wherein the second expected value generation circuit includes a second encoder circuit which generates a signal value corresponding to the number of the plurality of second selectors, and wherein the second detection circuit includes a second decoder circuit which receives signals from the plurality of fourth electrode and generates a value based on the signals from the plurality of fourth electrodes.
12. The semiconductor device according to claim 7, wherein the second semiconductor chip further includes at least one combinational logic circuit, and wherein in at least one second selector of the plurality of second selectors, a signal from at least one second flip-flop circuit of the plurality of second flip-flop circuits is input via the combinational logic circuit of the second semiconductor chip.
13. A semiconductor device testing method for a semiconductor device including: a first semiconductor chip which includes a first internal circuit, a plurality of first flip-flop circuits connected to the first internal circuit, a plurality of first selectors, and a plurality of first electrodes connected to respective outputs of the plurality of first selectors; a plurality of first connection conductors; and a second semiconductor chip which includes a plurality of second electrodes respectively connected to the plurality of first electrodes via the plurality of first connection conductors and a second internal circuit connected to at least one of the plurality of second electrodes, wherein at least one of the first semiconductor chip and the second semiconductor chip includes a part of a test circuit unit, wherein the test circuit unit includes a first detection circuit which receives a signal from each of the plurality of second electrodes, a first selector control circuit which controls the plurality of first selectors, a first expected value generation circuit which

generates a first expected value signal including a first expected value, and a test circuit which receives an output of the first detection circuit, and wherein the first expected value generation circuit generates and supplies the first expected value signal of the same first expected value to all of the plurality of first selectors, the first selector control circuit controls the plurality of first selectors to output the first expected value signal input to a first expected value input of each of the plurality of first selectors, the first detection circuit generates a value indicating a first detection result based on a signal input from each of the plurality of second electrodes, and the test circuit tests whether or not the first expected value matches a value indicating the first detection result.
14. The semiconductor device testing method according to claim 13, wherein the first expected value generation circuit generates and supplies the first expected value signal of the first expected value to the plurality of first selectors such that a first value of the first expected value of the first expected value signal is supplied to one of the plurality of first selectors and a second value of the first expected value of the first expected value signal different from the first value is supplied to other first selectors of the plurality of first sectors, the first selector control circuit controls the plurality of first selectors to output the first expected value signal input to the first expected value input of each of the plurality of first selectors, the first detection circuit generates a value indicating a first detection result based on signals input from the plurality of second electrodes, and the test circuit tests whether or not a set of the first expected values match a set of values indicating the first detection result.